Om Malviya

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Education

Birla Institute of Technology And Science, Pilani (BITS Pilani) **July 2024 – July 2027**
Bachelors in Computer Science *CGPA - 8.5*

Achievements

- **Expert @Codeforces** | Max Rating 1747 | **5* @CodeChef** | Max Rating 2069.
- Secured All World Ranks **1777** in **Meta Hacker Cup Round 1** (2025).
- **4 Kyu @Atcoder** | Max Rating 1258 | **Knight @Leetcode** | Max Rating 1959.
- Achieved **All India Rank 42** in **IIT Madras** Shaastra Programming Contest(Offline).
- 2× Podium Finishes in **National Hackathons** – **1st Place** (Algorand) | **3rd Place** (RIFT'26 – PW).

Projects

Limit Order Book | C++ Matching Engine **[GitHub]**

- Built a two-sided **limit order book** matching engine in C++20 with **price-time (FIFO) priority** for limit and market orders – modeling the core of an electronic exchange and the market-microstructure logic that underpins electronic trading.
- Modeled prices as **integer ticks** for exact, deterministic matching and **O(1)** depth-of-book volume reads; separated the zero-I/O matching core from rendering via emitted **trade events**.
- Validated with **19 Catch2 invariant tests** (no crossed book, quantity conservation, price-time priority), plus a CLI with a live depth ladder.

Search Typeahead | Low-Latency Distributed Backend (C++20) **[GitHub]**

- Engineered a low-latency search-suggestion service over a real **300k+ row Wikipedia Pageviews** dataset: a hand-written Trie serves ranked top-10 prefix matches, fronted by a distributed Redis cache sharded across 3 nodes via a consistent-hash ring.
- Cut database write pressure with a batched writer (one bulk UPSERT per flush) and added a recency-aware ranking model; benchmarked p50/p95/p99 latency and cache hit rate.
- **Stack:** C++20, PostgreSQL, Redis (hiredis), cpp-httplib, React/Vite frontend.

Website Automation Agent | Autonomous Browser Agent **[GitHub]**

- Engineered an autonomous agentic system modeled on **browser-use architecture** that executes multi-step web tasks in plain English via a real Chromium browser.
- Implemented interactive DOM element detection (**buildDomTree.js**) mapping bounding boxes to index maps, enabling structured text-based reasoning and dynamic tool execution without vision model latency/cost overhead.
- Built resilient execution loops featuring **Pydantic schema validation**, rate-limit backoff, stale-element recovery, and real-time **SSE streaming** via a FastAPI Web-UI.

Experience

Scaler - Internal Tools **March 2025 – June 2025**
Backend Developer / Tech Stack: Spring Boot, REST APIs, PostgreSQL *Bangalore, Karnataka*

- Contributed to the development of a **college management application** for Scaler, collaborating with a team to streamline campus operations. *(Project currently under development)*
- Implemented mess management system with database schema design **removing dependency on Google Forms**.
- Built announcement system with RESTful APIs that organizes communications in one place rather than scattered across WhatsApp groups, making announcements **searchable and accessible with 80% faster information retrieval**.
- Developed **15+ features** including transport management, document hub, events directory, digital ID card using Spring Boot and PostgreSQL, **addressing student and college pain points** through scalable architecture.

Technical Skills

Programming: Python, C++ (C++20), Java, JavaScript, SQL, TypeScript
AI & Agentic Systems: Agentic Workflows, Tool Calling, RAG / Vector Search, Structured Outputs (Pydantic), LLM Integration, Playwright, FastAPI
Backend & Systems: PostgreSQL, Redis, Spring Boot, Git, GitHub, Docker, Latency/Performance Benchmarking

Extracurricular

Ex- Core Member at NlogN – The Competitive Programming Club: Organized 5+ programming contests (AlgoStrick, Code Against Clock) with **500+ participants nationwide** and conducted CP sessions and contests.
HOD Sponsorships @ Ascent'26: Secured sponsorships worth **4.5L+ in total**, closing brands including Google Developer Groups, Claude Code Community London, and Red Bull.